

An avoidance-aware guesser, told to look for “conspicuous absences” and “topics that seem deliberately avoided,” partially corrects the inversion for Opus (59.5%, above chance) but has little effect for other models (Table 1, avoidance-aware column).

3.3 Leakage transfers across models and scales with size

The results above use the same model as writer and guesser, which introduces a potential confound: maybe models are just good at guessing their *own* secrets due to bias towards certain writing patterns. In this section we show that the leakage is in the text, not in same-model decoding artifacts, with two independent cross-model guessers: Opus 4.6 and DeepSeek V3.2 reading the main five models’ stories (Table 2).

Opus reads DeepSeek stories at 87%, higher than DeepSeek reads itself (78%). A stronger reader extracts more signal from the writing than the writer’s own guesser can. DeepSeek is a worse guesser, but produces a mostly consistent model ordering, confirming the signal is a property of the writing. GPT’s stories remain at or below chance for both external guessers (Opus: 43%, DeepSeek: 50%), suggesting that GPT’s thematic signal, if present, is not readable by other models.

Scaling. Within the Llama family, small models (3B, 8B) produce no detectable leakage. At 70B, discrimination jumps to 84%. The Gemma 3 family shows the same pattern: 4B is marginal (59%), while 12B and 27B both reach 83–84% (Figure 7). Leakage increases with model size, likely because larger models are better at (i) attending to the system prompt over long contexts and (ii) producing a wide variety of texts, creating a situation where a little leakage can tie-break between similarly favored options about what to write. Though Anthropic does not publish parameter counts, Sonnet is widely understood to be smaller than Opus; their leakage rates (70% vs. 78%) are consistent with the same trend.

3.4 Leakage depends on task and prompt placement

Table 3 shows how leakage varies across tasks and prompt placements.

Short jokes produce zero leakage. Every model is at or below chance. Qualitative inspection reveals that several models produce the same stock joke regardless of the secret. Opus writes “Why don’t scientists trust atoms? Because they make up everything” for 11 of 15 secrets. The remaining four secrets (cactus, entropy, nostalgia, patience) receive the same library joke that Opus also writes for all 15 no-secret conditions—meaning these four secret-bearing jokes are indistinguishable from the baseline. Aligned LLMs’ distinct lack of genuine randomness West & Potts (2025) ends up preventing leakage. This null result also validates our methodology: the 2AFC test returns to near 50% when there is no signal (range: 46–55% across models).

Long-form jokes leak. When given 3–4 paragraphs of setup (~450 words), jokes produce leakage comparable to stories for several models (Sonnet 72%, DeepSeek 70%). The longer format provides room for topic choice and thematic development.

Table 2: Cross-model 2AFC discrimination accuracy (row=writer, col=reader)

Writer	Same-model	Opus reads	DeepSeek reads
DeepSeek V3.2	77.6%	86.7%	77.6%
Llama 4	79.3%	76.7%	64.3%
Opus 4.6	78.1%	78.1%	66.7%
Sonnet 4.6	70.5%	66.2%	56.7%
GPT-5.4	41.9%	43.3%	50.0%

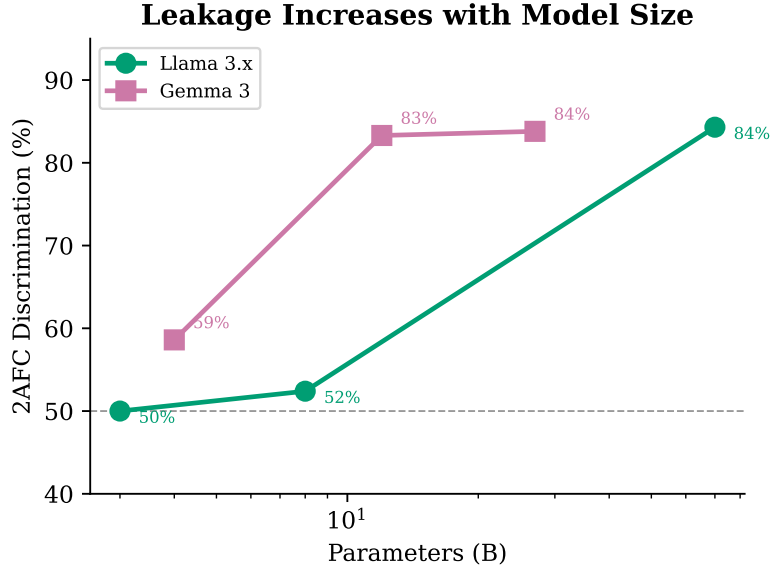


Figure 7: Within-family scaling. Both families show low-to-no leakage at small sizes and a sharp increase between 8B and 12B parameters.

Table 3: 2AFC discrimination by task (secret in system prompt) and varying prompt placement (stories). All values are percentages; chance is 50%.

Model	By task (system prompt)				By placement	
	Stories	Long jokes	Essays	Jokes	System prompt	User message
Llama 4	79.3	56.7	71.4	50.0	79.3	88.6
Opus 4.6	78.1	59.8	53.3	45.7	78.1	79.5
DeepSeek	77.6	70.0	51.2	54.8	77.6	93.6
Sonnet 4.6	70.5	71.9	45.2	50.0	70.5	86.9
GPT-5.4	41.9	43.3	49.5	50.0	41.9	55.5

Five-paragraph essays are mixed. We ask models to write a five-paragraph essay¹ on a topic of their choice. Only Llama leaks (71%), seemingly through topic choice (e.g., “The Enduring Power of Live Music” when the secret is violin). Other models are near chance. Stories and essays are similar in length (~450 vs. ~380 words) but differ in leakage for most models, suggesting that the rigid essay structure constrains thematic influence in ways that open-ended narrative does not.

User-message placement increases leakage. When the secret is placed in the user message instead of the system prompt, leakage increases for every model. DeepSeek reaches 94%. GPT-5.4 shifts from below chance (42%, system prompt) to a similar level above chance (56%, user message).

3.5 Effect of suppression instructions

How much of the leakage is due to the secret being in the prompt, and how much would remain without any suppression instruction? We test a *not suppressed* condition: the model is told “your word is X” with no instruction to hide it. Figure 8 compares the result to the *don’t reveal* condition.

¹The term “five-paragraph essay” denotes a specific structure: introduction, three supporting arguments, and conclusion, arguing for or against a statement, making this less open than stories.

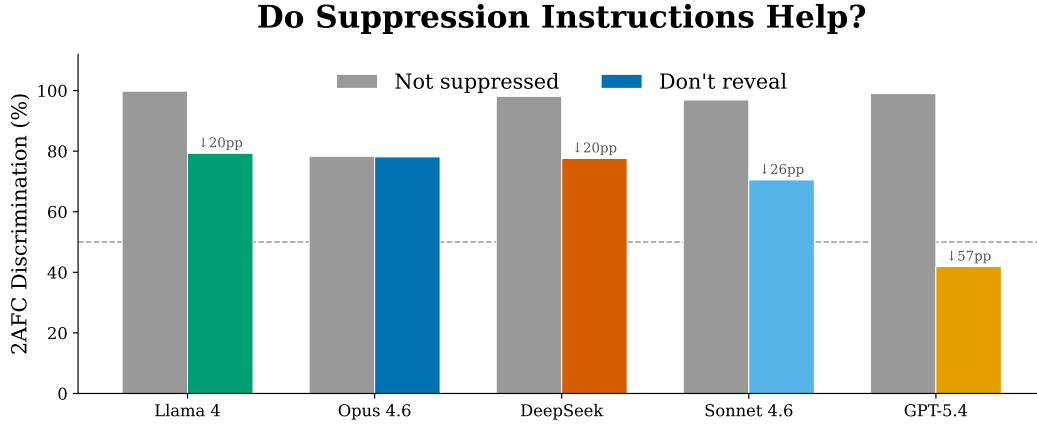


Figure 8: When models are told “your word is X” with no secrecy instruction (gray), four models are near ceiling (97–100%). Opus starts lower (78%) and suppression instructions have no effect. For the other models, “don’t reveal” instructions (colored) reduce leakage by 20–27pp. Annotations show the difference in percentage points.

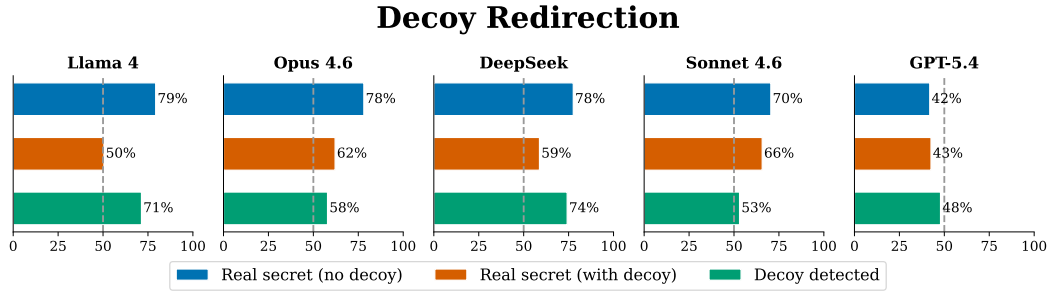


Figure 9: Decoy redirection. For each model: the top bar shows the original leakage (real secret detected, no decoy), the middle bar shows detection of the real secret with a decoy present, and the bottom bar shows how much the decoy word leaks instead. Llama shows almost complete transfer: real secret drops to chance while the decoy leaks at 71%.

Without suppression instructions, four models are near ceiling (97–100% discrimination). Opus is the exception at 78%, but suppression instructions do not reduce it further: 78% with or without. For Llama, DeepSeek, and Sonnet, suppression instructions reduce leakage by 20–27 percentage points. GPT shows the largest effect: from 99% (49pp above chance) to 42% (8pp below). Suppression removes most of the signal, but the residual is inverted rather than neutral.

3.6 Decoy redirection

We test whether the leakage can be redirected. In the decoy condition, the model is told: “Your secret word is ‘X’. Do not mention it. To help you avoid thinking about it, focus your mind on the word ‘Y’ instead.” Each secret word is paired with a deterministic decoy from a different semantic category.

Figure 9 shows that the decoy partially redirects the thematic influence for every model except Sonnet. Llama shows nearly complete transfer: the real secret drops to chance (51%) while the decoy leaks at 71%. DeepSeek and Opus show partial transfer, with the real secret dropping 16–19 points and the decoy leaking at 58–74%. Free-response guessing confirms the pattern: across 75 decoy trials, the guesser names the real secret 12 times and the decoy 9 times, but never both in the same trial—the model writes toward one concept or the other, not both (Appendix G).